

Comp683: Computational Biology

Lecture19

April 13, 2026

Today + Announcements

- Finishing pp integrating multiple relational definitions between genes or proteins
- Multiomics Factor Analysis (MOFA)
- Homework 2 is due Wednesday
- Journal club Monday

Defining Consensus d -dimensional vectors for nodes

Given multiple relational definitions (e.g. multiple graphs) between a common set of nodes (features), define a consensus d -dimensional embedding vector for each node that aligns well with each individual graph (e.g. distinct relational definitions).

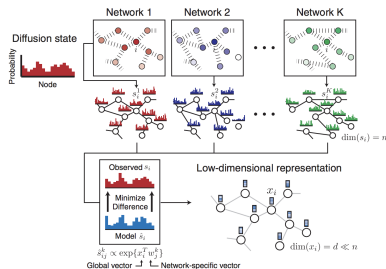


Figure: from Cho *et al.* Cell Systems. Each graph is representing a different relational definition between node (features).

Review: Random walk with re-start

$$s_i^{t+1} = (1 - p_r)Bs_i^t + p_re_i$$

- The first term corresponds to following a random edge connected to the current node
- The second term corresponds to restarting from node i .
- At some point, this reaches a stationary distribution, s_i^∞ , or fixed point
- When the diffusion states between two nodes are close, this implies they have similar positions in the graph with respect to other nodes.

Quantifying Topological Overlap Between a Node Pair

Each node is given two vector representations, $\mathbf{w}_i, \mathbf{x}_i \in \mathbb{R}^d$

- Let $\mathbf{w}_i$ refer to the context feature of a node (e.g. per relational definition)
- Let $\mathbf{x}_i$ refer to the node feature of node i (e.g. overall)

Define a new similarity measure between nodes i and j as,

$$\hat{s}_{ij} = \frac{\exp\{\mathbf{x}_i^T \mathbf{w}_j\}}{\sum_{j'} \exp\{\mathbf{x}_i^T \mathbf{w}_{j'}\}}$$

Unpacking

$$\hat{s}_{ij} = \frac{\exp\{\mathbf{x}_i^T \mathbf{w}_j\}}{\sum_{j'} \exp\{\mathbf{x}_i^T \mathbf{w}_{j'}\}}$$

- If $\mathbf{x}_i$ and $\mathbf{w}_j$ are close in direction and hence have a large inner product, then node j should be frequently visited in the random walk starting from node i .

Recap of what is happening

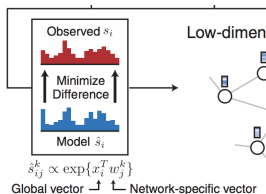


Figure: from Fig. 1. Given observed diffusion states from RWR, we should be able to find a global vector ($\mathbf{x}$) and view-specific vector ($\mathbf{w}$), such that a function of $\mathbf{x}$ and $\mathbf{w}$ gives a good diffusion state approximation.

Writing Out an Objective Function for Embeddings

To find optimal d -dimensional representations for each node, formulate an optimization problem that minimizes the notation between s (RWR diffusion state) and $\hat{s}$ ('approximation') as,

$$\underset{w,x}{\text{minimize}} C(s, \hat{s}) = \frac{1}{n} \sum_{i=1}^n D_{KL}(s_i \| \hat{s}_i)$$

Written out, given our definition of $\hat{s}$ gives the following (with $H(\cdot)$ denoting entropy),

$$C(s, \hat{s}) = \frac{1}{n} \sum_{i=1}^n \left[-H(s_i) - \sum_{j=1}^n s_{ij} \left(x_i^T w_j - \log \left(\sum_{j'=1}^n \exp \{ x_i^T w_{j'} \} \right) \right) \right]$$

Integrating Heterogeneous Networks

You can do these RWRs on each individual network. At the same time, you can let x be fixed across all relational definitions.

Similar to what we have seen, yet adapted for modality k , we can write,

$$\hat{s}_{ij}^k := \frac{\exp \left\{ x_i^T w_j^k \right\}}{\sum_{j'} \exp \left\{ x_i^T w_{j'}^k \right\}}$$

Writing the Objective Function Across All Modalities

Now, the objective function can be rewritten to take into account the recently-computed $\hat{s}_{ij}^k$ s, and sums over all modalities as,

$$\underset{w,x}{\text{minimize}} C(s, \hat{s}) = \frac{1}{n} \sum_{k=1}^k \sum_{i=1}^n D_{KL} \left(s_i^k \| \hat{s}_i^k \right)$$

Implementation (the slow way)

To find the optimal w s and x s for each node, you could compute gradients, which turn out to be,

$$\nabla_{w_i^k} C(s, \hat{s}) = \frac{1}{n} \sum_{j=1}^n \left(\hat{s}_{ji}^k - s_{ji}^k \right) x_j$$

and

$$\nabla_{x_i} C(s, \hat{s}) = \frac{1}{n} \sum_{k=1}^K \sum_{j=1}^n \left(\hat{s}_{ij}^k - s_{ij}^k \right) w_j^k$$

An SVD Formulation: Setup

- Let S^k be the $N \times N$ diffusion state matrix for network k
- Also, let s_i^k be the i th column of this matrix, S^k

These matrices can be concatenated to form an $NK \times N$ matrix, S

Remember Truncated SVD?

The authors used truncated SVD as an alternative to estimating the w_i^k s and x_i s as,

$$S = U\Sigma V$$

(Remember this implies that we will have some 0s on the diagonal (e.g. zeroed out singular values) of Σ)

- $\{w_i^k\} \rightarrow \Sigma^{1/2} U^T \rightarrow (d \times d) \times (d \times NK)$
- $\{x_i\} \rightarrow \Sigma^{1/2} V \rightarrow (d \times d) \times (d \times N)$

Using the Learned $\mathbf{x}_i$ s as Feature Vectors

- After Mashup each node, i has an embedding, $\mathbf{x}_i$.
- Each protein has a known function, which we can try to predict.

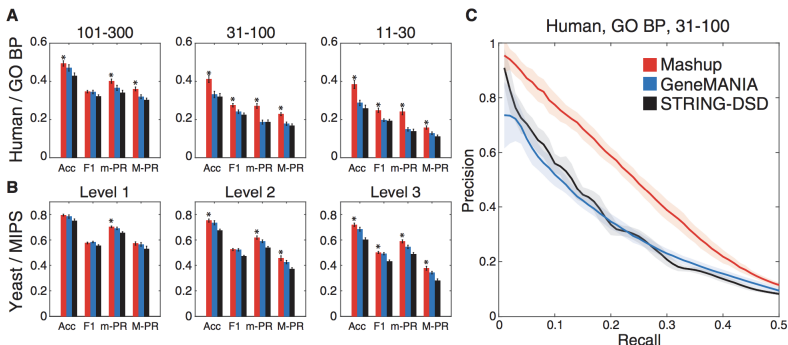


Figure: from Fig. 2. Performance is evaluated for multiple levels of annotation. Precision = True Pos/True Post + False Pos, Recall = True Pos/True Pos + False Neg

Similarly, Combining All Networks Leads to Better Protein Function Prediction

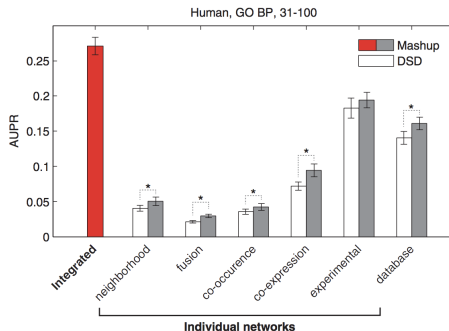


Figure: from Fig. 3. DSD is some single-graph diffusion method. Note that experimental is also a top performer.

Intuition about Parameters

The main parameters of interest is the restart probability, and the number of dimensions to keep.

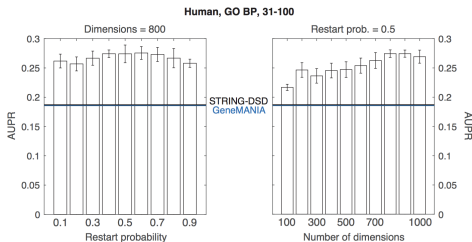


Figure: from Supp Fig. 4

*Btw, I recommend choosing your y-axis so that it is useful when making such a plot.

Mashup is More Robust to Noise in the Network

Here, noise was simulated by removing a subset of edges from the original graph.

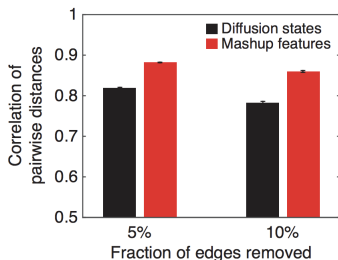


Figure: from Supp. Fig. 7. Edges were removed from the BioGrid physical interaction network. Similarities between nodes could be calculated based on diffusion state or mashup.

Recap

- Context-specific relational information and general relational information are used to construct a joint representation for each node, based on accurate modeling of diffusion states.
- Vector representation can be used to better predict node functions.

Switching gears to another multiomics data integration strategy: MOFA

Notation and Goals

- Start with M data matrices, $\mathbf{Y}^1, \mathbf{Y}^2, \dots, \mathbf{Y}^M$ on a set of N samples
- A Particular modality's matrix, $\mathbf{Y}_m$ has dimensions, $N \times D_m$

The goal is to write the input data matrix, $\mathbf{Y}^m$ as a product of a common factor matrix, $\mathbf{Z}$ (samples $\times$ factors) and a 'weight' matrix (e.g. one that relates features to factor space) as,

$$\mathbf{Y}^m = \mathbf{Z}\mathbf{W}^{mT} + \epsilon^m$$

Unpacking...

Assuming there are k factors and given

$$\mathbf{Y}^m = \mathbf{Z}\mathbf{W}^{mT} + \epsilon^m$$

- $\mathbf{Z} \in \mathbb{R}^{N \times K}$ relate the original samples to the factors
- $\mathbf{W}^m \in \mathbb{R}^{D_m \times K}$ relates the original features to the factors

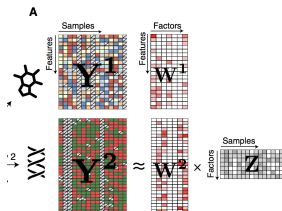


Figure: from Argelaguet *et al.* Molecular Systems Biology. 2018.

Overview of the Model

This is latent variable model and feature dependencies are attempted to be explained in terms of k latent classes (or 'factors').

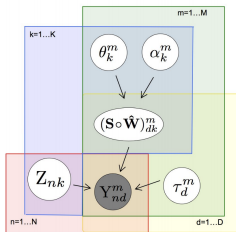


Figure: from Argelaguet *et al.* Molecular Systems Biology. 2018. As always, gray nodes are observed variables and white nodes are the unobserved variables inferred by the model.

Starts Simple Enough....

The the observed value of feature d in sample n from modality m is modeled as,

$$p(y_{nd}^m) = \mathcal{N}(y_{nd}^m \mid \mathbf{z}_{n,:} \mathbf{w}_{d,:}^{mT}, 1/\tau_d^m)$$

- $\mathbf{w}_{d,:}^m$: gives the d -th row of $\mathbf{W}^m$
- $\mathbf{z}_{n,:}$: is the n -th row of the latent factor matrix, $\mathbf{Z}$.
- **Other priors:**
 - $p(z_{n,k}) = \mathcal{N}(z_{n,k} \mid 0, 1)$
 - $p(\tau_d^m) = \mathcal{G}(\tau_d^m \mid a_0^\tau, b_0^\tau)$

Regularization in $\mathbf{W}^m$

In computing each $\mathbf{W}^m$ there two desired kinds of sparsity :

- View and factor-wise sparsity
- Feature-wise sparsity

The intuition is that $\mathbf{w}_{:,k}^m$ is shrunk to 0 if factor k does not drive any variation in view m .

In practice...

Sparsity is enforced for each $\mathbf{W}$ through appropriate priors. Specifically, they model $\mathbf{W}$ as a product of a Gaussian random variable, $\hat{w}$, and a Bernoulli random variable, s .

$$p(\hat{w}_{d,k}^m, s_{d,k}^m) = \mathcal{N}(\hat{w}_{d,k}^m \mid 0, 1/\alpha_k^m) \text{Ber}(s_{d,k}^m \mid \theta_k^m)$$

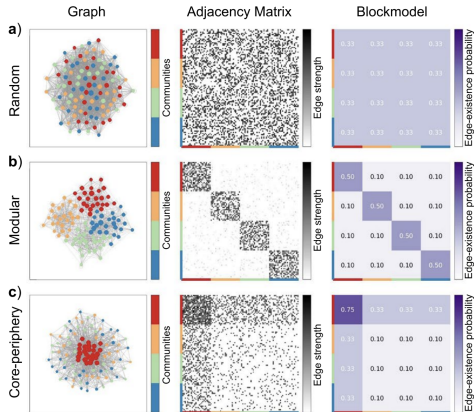
- $p(\theta_k^m) = \text{Beta}(\theta_k^m \mid a_0^\theta, b_0^\theta)$
- $p(\alpha_k^m) = \mathcal{G}(\alpha_k^m \mid a_0^\alpha, b_0^\alpha)$

Thinking More About Sparsity

Let's think about what happens with bernoulli probabilities:

- A value of θ_k^m close to 0 implies that most of the weights of factor k in view m tend towards 0.
- A θ_k^m close to 1 alternatively implies that most of the weights are non-zero (e.g. a non-sparse factor)

Living in the world of latent variables... remember stochastic block model?



Latent variable: node-to-community assignments

$$\begin{aligned}\log \mathcal{L}(\mathcal{X}, \mathcal{Z}) = & \sum_i \sum_q Z_{iq} \log \alpha_q \\ & + \frac{1}{2} \sum_{i \neq j} \sum_{q, \ell} Z_{iq} Z_{j\ell} \log b(X_{ij}; \pi_{q\ell})\end{aligned}$$

- At some point in updating parameters via EM, we need to worry about the posterior or $P(\mathcal{Z} \mid \mathcal{X})$.

Dealing with Latent Variables

More help with factorized approximations...

- Our situation is the case of having an intractable posterior distribution of unobserved variables $p(\mathbf{X} \mid \mathbf{Y})$.
- Here, $\mathbf{X}$ is denoting our un-observed variables, and $\mathbf{Y}$ is denoting the observed variables

We can deal with $p(\mathbf{X} \mid \mathbf{Y})$ by approximating it with a factorized form,

$$q(\mathbf{X}) = \prod_i q(\mathbf{X}_i)$$

.

This can make inference more efficient.

Reminder About Latent Variables Here

Notably, $\mathbf{Z}$ and $(\mathbf{S} \circ \hat{\mathbf{W}})$

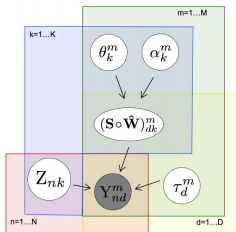


Figure: from Argelaguet *et al.* Molecular Systems Biology. 2018. As always, gray nodes are observed variables and white nodes are the unobserved variables inferred by the model.

For MOFA : Mean Field Approximation

The assumption under the mean-field approach is that $q(\mathbf{X})$ factorized over M disjoint groups of variables as,

$$q(\mathbf{X}) = \prod_{i=1}^M q(\mathbf{x}_i)$$

.
In the case of MOFA,

$$\begin{aligned} q(\mathbf{Z}, \mathbf{S}, \hat{\mathbf{W}}, \boldsymbol{\alpha}, \boldsymbol{\tau}, \boldsymbol{\theta}) &= q(\mathbf{Z})q(\boldsymbol{\alpha})q(\boldsymbol{\theta})q(\boldsymbol{\tau})q(\mathbf{S}, \hat{\mathbf{W}}) \\ &= \prod_{n=1}^N \prod_{k=1}^K q(z_{n,k}) \prod_{m=1}^M \prod_{k=1}^K q(\alpha_k^m) q(\theta_k^m) \prod_{m=1}^M \prod_{d=1}^{D_m} q(\tau_d^m) \prod_{m=1}^M \prod_{d=1}^{D_m} \prod_{k=1}^K q(\hat{w}_{d,k}^m, s_{d,k}^m) \end{aligned}$$

Moving into Results: Summary

After having inferred $\mathbf{W}^m$ s and $\mathbf{Z}$, several things can be done, such as, imputation, annotation of factors, etc.

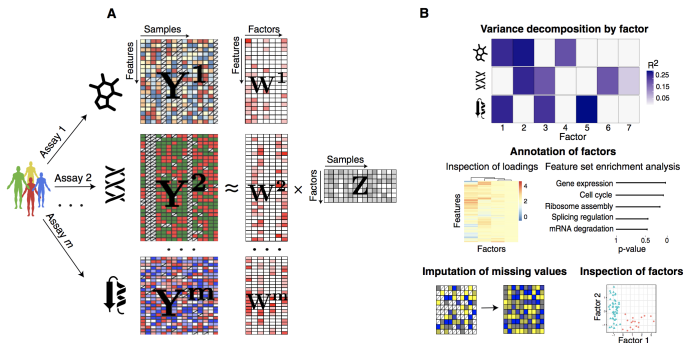


Figure: from Argelaguet *et al.* Molecular Systems Biology. 2018.

Getting Intuition About Variance Explained by Factors

- **Variance Explained by Factor k in View m** : Compute the fraction of variance explained by factor k in view m as,

$$R_{m,k}^2 = 1 - \left(\sum_{n,d} y_{nd}^m - z_{nk} w_{kd}^m - \mu_d^m \right)^2 / \left(\sum_{n,d} y_{nd}^m - \mu_d^m \right)^2$$

Here, μ_d is the feature-wise mean.

Variance Explained per Modality

- **Variance Explained In Each Modality Over All Features :**

Compute the fraction of variance explained per view as,

$$R_m^2 = 1 - \left(\sum_{n,d} y_{nd}^m - \sum_k z_{nk} w_{kd}^m - \mu_d^m \right)^2 / \left(\sum_{n,d} y_{nd}^m - \mu_d^m \right)^2$$

Here, μ_d^m is the feature-wise mean.

Results in Dataset 1 : CLL dataset

Find data here

https://rdrr.io/github/bioFAM/MOFAdata/man/CLL_data.html

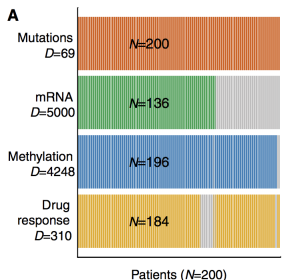


Figure: from Argelaguet *et al.* Molecular Systems Biology. 2018. Modalities and present/missing features for each patient.

Visualization of Samples

Samples can be projected into two dimensions based on the first two factors inferred in the matrix, $\mathbf{Z}$

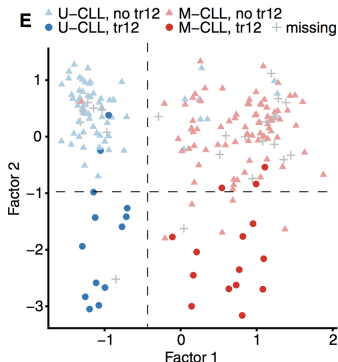


Figure: colors denote something related to the status of the tumor. Shape indicates chromosome 12 trisomy status.

Jointly Predicting Clinical Outcome in Contrast to a Single Type of Data

Experiment :

- Predict time to treatment for $N = 174$ patients using multivariate Cox regression trained using the 10 factors from MOFA
- Compare this performance on the prediction accuracy on individual modalities.
 - In this case, reduce each individual modalities to the top 10 PCs.

Results Predicting Time to Treatment

Note that the Y-axis is simply a statistic reflecting goodness of fit between true and predicted times to treatment (Use top PCs instead of regular feature space).

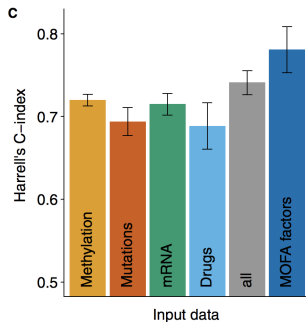


Figure: from Fig. 4 of Argelaguet *et al.* Molecular Systems Biology. 2018.

Instead of using PC Representations for Each Modality

Instead of using PC representations for each modality, the authors also compared to prediction with all features from each individual modality. Again, predicting time to treatment.

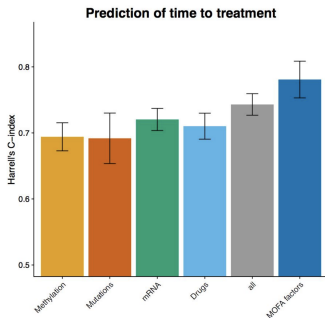


Figure: from Supp. Figure 20. Using all features still does not beat performance based on MOFA factors.

Certain Combinations of Modalities Can Turn out to be More Effective

Applying MOFA with individual modalities held out reveals sometimes certain combinations of modalities is more effective than just throwing everything in all together!

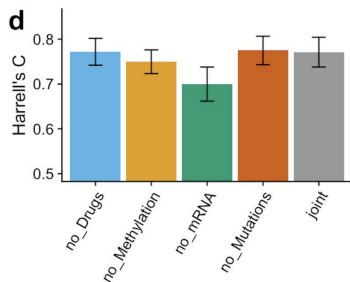


Figure: from Supplementary Figure 8.

Survival Analysis According to Particular Factors

Survival probability vs time-to-treatment for the individual MOFA factors

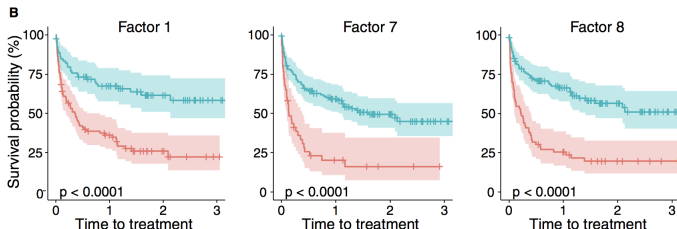


Figure: from Fig. 4. We can see distinct patterns based on survival probability against predicted time to next treatment.

Dataset 2 : Single-Cell Data for a Differentiation Trajectory

Data from a single-cell multiomics dataset. This is applied to 87 mouse embryonic stem cells, with 16 that were cultured in '2i1' media, giving them a naive pluripotent state. The remaining cells were serum grown and hence committed to a differentiation trajectory.

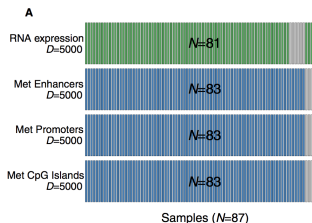


Figure: from Fig. 5. Distribution of samples profiled in each modality.

Visualizing Cells Based on $\mathbf{Z}$

Using $\mathbf{Z}$ to plot cells based on the first two factors, the cells separate according to how they were cultured (which also corresponds to differentiation status!)

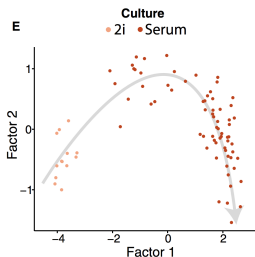


Figure: from Fig. 5.

Inferred Factors on Genes and How They Relate to Pluripotency and Differentiation

Using the $\mathbf{W}^m$ s for the mRNA data, the authors compared the ranks of genes involved in pluripotency and differentiation, respectively to the loadings.

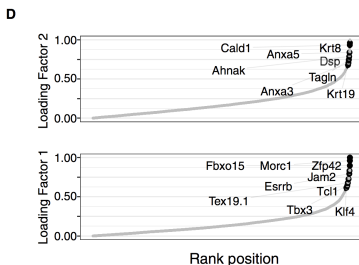


Figure: from Fig. 5. Pluripotency genes (top) vs differentiation related genes (bottom)

An Extension, MOFA+

A trivial extension....almost the same, except now we learn a $\mathbf{Z}$ per group.

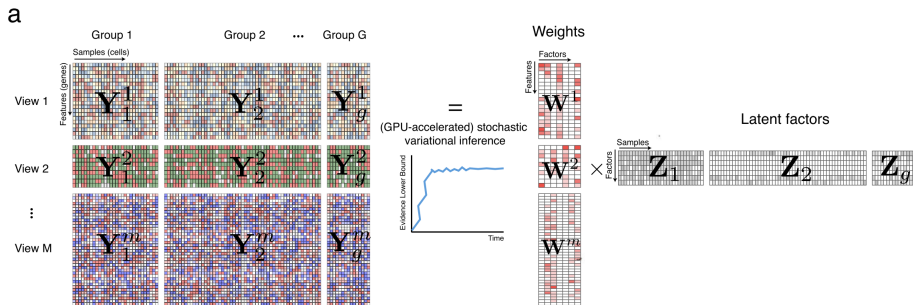


Figure: from Argalaguet *et al.* Genome Biology. 2020. Views are modalities. Groups are some partitioning of the samples.

So what do we think? Does anyone have any ideas for when this would be useful for your application?

Defining Y_{gm}

Similar to what we have seen with the regular MOFA,

$$Y_{gm} = Z_g W_m^T + \epsilon_{gm}$$

- Y_{gm} is the matrix of observations for the m th modality and g th group
- W_m is the weight matrix for the m th modality
- Z_g is the factor matrix for the g th group
- ϵ_{gm} is the residual noise for the m th modality in the g th group

Summary

- MOFA for decomposing a sample $\times$ feature matrix to (feature $\times$ factor)(factor $\times$ sample)
- MOFA+ for doing this calculation per group